

2. Further coordinate systems		
2.1	Cartesian and parametric equations for the ellipse and hyperbola.	Extension of work from FP1. Students should be familiar with the equations: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1; x = a \cos t, y = b \sin t.$ $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1; x = a \sec t, y = b \tan t;$ $x = a \cosh t, y = b \sinh t.$
2.2	The focus-directrix properties of the ellipse and hyperbola, including the eccentricity.	For example, students should know that, for the ellipse, $b^2 = a^2(1 - e^2)$, the foci are $(ae, 0)$ and $(-ae, 0)$ and the equations of the directrices are $x = +\frac{a}{e} \text{ and } x = -\frac{a}{e}.$
2.3	Tangents and normals to these curves.	The condition for $y = mx + c$ to be a tangent to these curves is expected to be known.
2.4	Simple loci problems.	